PROJECT NAME

Integrated Decision-Support System for Food Surplus and Waste Management in Palestine

Abstract

Food waste is both a disposal problem and a resource-efficiency problem: when food is discarded, the water, energy, land, labor, and transport used to produce it are also wasted. This paper proposes an institution-level decision-support system for Palestinian food-service operations. The system combines demand forecasting to reduce avoidable overproduction with batch-level optimization for unavoidable surplus. Each food batch is evaluated against food-safety status, remaining usable time, recipient demand, transport feasibility, and locally available recovery pathways. The proposed MVP is evaluated through a simulated restaurant and stress-tested under changing demand, recipient availability, and capacity. The research contribution is not a new marketplace or composting technology, but an integrated decision layer that links prevention, redistribution, and organic-resource recovery within one operational framework.

Keywords: food waste; surplus food; decision support; optimization; circular economy; Palestine

1. Problem and Environmental Significance

The United Nations Environment Programme estimates that 1.05 billion tonnes of food were wasted in 2022 and that food loss and waste account for approximately 8-10% of global greenhouse-gas emissions [1]. The environmental burden extends beyond methane released during disposal: wasted food also embodies unnecessary water use, energy consumption, agricultural land, packaging, and transport. This is especially relevant in Palestine, where resource efficiency and waste-management capacity are important environmental constraints. A 2023 study of municipal solid waste in Nablus found that bio- and organic material represented about 68% of the sampled waste stream; this figure is not equivalent to edible food waste, but it demonstrates the scale of recoverable organic material [2]. At the institutional level, An-Najah National University reported 3.8 tonnes of food waste during the 2024-2025 academic year and already measures waste daily across food-service locations [3]. These data support a shift from simple disposal management toward prevention and higher-value recovery.

2. Existing Solutions and Research Gap

Existing systems solve important but different parts of the problem. Winnow uses AI-enabled kitchen monitoring and forecasting to reduce overproduction [4]. Too Good To Go now provides an end-to-end retail surplus platform covering near-expiry inventory, markdowns, marketplace sales, and charity redistribution [5]. In Palestine, Laheq Halak connects vendors with consumers for discounted surplus or near-expiry food and includes donation functionality [6]. These solutions show that another marketplace alone would add limited novelty. The proposed research gap is narrower and operational: food-service institutions still need a local decision layer that connects prevention with batch-by-batch allocation across both human-use and organic-recovery pathways, while reacting to changing time, demand, transport, and capacity constraints. The U.S. EPA Wasted Food Scale provides an environmental hierarchy for this logic, prioritizing prevention and human use over lower-value disposal pathways [7].

Comparative positioning.

System	Prevention / forecasting	Sale / donation	Organic-recovery decision	Dynamic re-optimization
Winnow	Yes	No	No	No
Too Good To Go	Retail expiry / markdowns	Yes	No	Not across recovery paths
Laheq Halak	No	Yes	No	No
Proposed system	Yes	Yes	Yes	Yes

3. Proposed Decision-Support Framework

The proposed system operates in two linked stages. First, historical preparation and consumption data are used to estimate near-term demand and reduce avoidable overproduction. Second, unavoidable surplus is represented as individual food batches defined by quantity, category, human-consumption approval, remaining usable time, storage condition, and economic value. Candidate destinations may include discounted sale, donation, redistribution, composting, or energy recovery. Food safety is treated as a hard constraint rather than an AI prediction: a batch not approved for human consumption cannot be routed to sale or donation.

Optimization model.

Decision variable: x(i,j) = quantity of food batch i assigned to destination j.
Priority objective: maximize safe food retained for human use; then minimize disposal and logistics burden.
Core constraints: assigned quantity <= available quantity; destination capacity and demand; arrival before usable-time limit; transport/storage eligibility; food-safety eligibility.

The scientific basis combines established food-rescue optimization with adaptive allocation. Dubey and Tanksale model surplus-food recovery as a multi-depot vehicle-routing problem with time windows, split pickup, and split delivery [8]. Gao and Chen formulate uncertain food-bank allocation as a dynamic decision problem and propose an adaptive Re-Optimized (ReOPT) policy [9]. MIT Center for Transportation and Logistics further demonstrates the value of facility-specific

modeling when selecting circular food-waste pathways across catering operations [10]. The proposed system adapts these principles to a small, low-cost Palestinian food-service setting rather than reproducing any single existing platform.

4. MVP and Evaluation Methodology

The MVP will be tested using a simulated restaurant representing one realistic operating day. The dataset will include prepared, sold, and surplus quantities, together with virtual recipients and recovery facilities with defined demand, capacity, time windows, and transport constraints. The system will generate an allocation plan and an explanation for each decision. To test robustness, the simulation will introduce events such as a recipient becoming unavailable, a sudden demand change, or reduced transport capacity; the model must then re-optimize. Performance will be compared with a simple baseline policy such as assigning each batch to the first eligible destination. Simulated results will be presented explicitly as algorithm validation, not as claimed real-world environmental savings.

Evaluation indicators.

Surplus prevented	Edible rescue rate	Landfill diversion	Transport efficiency
kg / %	kg / %	kg / %	km per kg

5. Expected Environmental Impact and Scalability

The primary expected benefit is environmental: less overproduction means fewer embedded resources are wasted, while better routing of unavoidable surplus reduces organic material sent to disposal. The EPA identifies prevention, donation, and upcycling as environmentally preferred pathways and landfill as one of the least preferred pathways because food decomposition generates methane and valuable nutrients are lost [7]. The project will therefore measure impact through avoided surplus, food retained for human use, organic material diverted from disposal, and transport efficiency. Greenhouse-gas savings will only be reported when pathway-specific, defensible emission factors are available. The initial architecture requires no dedicated sensors or processing hardware, allowing deployment in one cafeteria or restaurant and later integration with universities, hospitals, hotels, charities, surplus marketplaces, composting operations, or energy-recovery facilities.

6. Scope and Limitations

The hackathon MVP will validate decision logic rather than claim field-proven environmental savings. Its first evaluation uses simulated restaurant data, so operational performance must later be validated with real cafeteria records, verified food-safety policies, actual recipient capacities, and local transport data. Emission reductions will not be reported as measured impact until credible local or pathway-specific factors are available. This separation between simulated validation and real-world impact is necessary to keep the results scientifically defensible.

References

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[9] Gao, X., & Chen, X. (2026). A Decision-Support Framework for Food Bank Operations: Overcoming Waste and Uncertainty in Food Distribution. Link

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